

Highlights

What Do LLMs Say That Human Facilitators Don't? A Computational Behavioral Analysis of AI vs. Human Collaborative Design Meeting Facilitation

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- Llama 3.1 8B Instruct and human Project Manager turns differ significantly on 4 of 6 behavioral dimensions
- Human Project Manager turns produce greater embedding-space contextual semantic distance
- Llama 3.1 generates topically divergent responses (higher Jensen–Shannon distance from context)
- Llama 3.1 facilitation behavior is largely temperature-stable across 0.3–1.0
- Findings provide a baseline comparison of AI vs. human PM facilitation turns in design meetings

What Do LLMs Say That Human Facilitators Don't? A Computational Behavioral Analysis of AI vs. Human Collaborative Design Meeting Facilitation

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Abstract

Artificial intelligence is increasingly explored for assisting collaborative design meeting facilitation, yet empirical comparisons between large language models (LLMs) and human facilitation turns remain limited. This study presents a computational comparison of naturally occurring human Project Manager facilitation turns and LLM-generated facilitation moves in matched collaborative design contexts. Analyzing 199 matched facilitation move pairs drawn from Project Manager turns in the AMI Meeting Corpus (Carletta et al., 2005), we compare human PM turns with Llama 3.1 8B Instruct (Dubey et al., 2024) across six behavioral dimensions: semantic distance from context, directiveness (exploratory classifier), linguistic specificity proxy, empathy-related lexical density, topic divergence (Jensen–Shannon distance), and phase-vocabulary alignment. Feature extraction used sentence-transformer embeddings (Reimers and Gurevych, 2019), an exploratory DistilBERT classifier (Sanh et al., 2019), and LDA-based topic modeling (Blei et al., 2003). Inter-rater reliability for facilitation move identification from raw transcripts was $\kappa = 0.81$. Mann-Whitney U tests with Bonferroni correction revealed significant differences on four of six dimensions ($p_{\text{Bonf}} < 0.05$). Human PM

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turns exhibited greater contextual semantic distance ($r = -0.54$), whereas Llama 3.1 8B Instruct generated responses with higher topic divergence ($r = +0.58$), higher specificity proxy ($r = +0.25$), and higher phase-vocabulary alignment ($r = +0.34$). No statistically significant differences were detected for directiveness ($r = -0.005$) or empathy-related lexical density ($r = -0.058$); both dimensions were underpowered (power = 0.05 and 0.21), and these null findings cannot establish equivalence. Decoding temperature (0.3–1.0) modestly affected specificity ($\rho = +0.12$, $p = 0.005$) and phase-vocabulary alignment ($\rho = -0.10$, $p = 0.013$); the remaining four dimensions were temperature-stable. These findings document empirical differences between Llama 3.1 8B Instruct and human Project Manager participants in scenario-based design meetings, establishing baseline behavioral characteristics for AI-assisted facilitation.

Keywords: collaborative design meetings, facilitation, large language models, Llama 3.1, behavioral analysis, AMI Meeting Corpus

1. Introduction

Collaborative design meetings are a primary site of creative problem-solving in industry, education, and public policy (Brown, 2008; Cross, 2011). Central to these collaborative processes is the role of the meeting facilitator, whose verbal interventions—asking questions, reframing problems, and prompting exploration—shape how teams navigate problem and solution spaces (Seidel and Fixson, 2013; Starostka et al., 2021). As instruction-tuned large language models (LLMs) demonstrate expanding capabilities in conversational interaction, researchers and practitioners are exploring their application as AI-powered facilitation assistants in online discussion (Korre et al., 2025) and collaborative design environments (Gmeiner et al., 2023). However, a key empirical question remains: how do the facilitation moves generated by an LLM differ from the naturally occurring facilitation moves produced by human facilitators in design contexts?

This question is relevant to behavioral and computational design research because facilitation moves serve as observable communicative actions reflecting underlying strategies, domain context, and collaborative dynamics. Schön (1983) conceptualized professional practice as “reflection-in-action,” in which facilitators draw on situated repertoires shaped through experience. From a Social Cognitive Theory perspective (Bandura, 1986), human

facilitators develop communicative behaviors through observation, feedback, and social interaction within authentic team environments. By contrast, current LLM text generation produces fluent outputs through statistical pattern matching over broad pre-training corpora, constrained by instruction-tuning and prompt templates, rather than situated social learning. The six behavioral dimensions evaluated in this study—contextual semantic distance, directiveness, linguistic specificity proxy, empathy-related lexical density, topic divergence (Jensen–Shannon distance), and phase-vocabulary alignment—provide computational proxies to compare these output characteristics. The AMI Meeting Corpus (Carletta et al., 2005), which captures scenario-based design team meetings led by designated Project Managers, offers an empirical dataset for analyzing naturally occurring facilitation turns.

Prior computational research on design discourse has examined coding schemes and protocol analysis (Kan and Gero, 2017; Stempfle and Badke-Schaub, 2002; Chandrasegaran et al., 2019), while studies of LLMs in creative contexts have investigated co-writing (Lee et al., 2022; Yuan et al., 2022) and ideation support (Yu et al., 2025). However, empirical comparisons isolating the specific behavioral properties of LLM-generated facilitation moves relative to human meeting turns remain limited.

This study presents a computational comparison of naturally occurring human Project Manager facilitation moves and LLM-generated facilitation moves in matched collaborative design contexts, addressing three specific research questions:

- **RQ1:** Do Llama 3.1 8B Instruct responses and human Project Manager turns differ significantly across the six behavioral dimensions?
- **RQ2:** Which dimensions show the largest behavioral differences, and in what direction?
- **RQ3:** How does decoding temperature affect Llama 3.1 8B Instruct facilitation behavior across these dimensions?

We evaluate these questions through a computational pipeline that extracts facilitation turns from the AMI Meeting Corpus (Carletta et al., 2005), generates matched responses using Llama 3.1 8B Instruct (Dubey et al., 2024) across three temperature settings, and evaluates behavioral differences using non-parametric statistical tests and sensitivity analyses.

55 2. Literature Review

2.1. Collaborative Design Meeting Facilitation

Collaborative design processes involve iterative phases of empathizing, defining, ideating, prototyping, and testing (Brown, 2008). Facilitators guide teams through these phases using verbal interventions such as open-ended
60 questions, reframings, and process prompts rather than direct decision-making (Seidel and Fixson, 2013). Starostka et al. (2021) presented a taxonomy of design facilitation moves, distinguishing between directive interventions (prescribing specific activities) and generative interventions (opening alternative perspectives). Dorst and Cross (2001) highlighted that design problem-
65 solution spaces co-evolve through interactive discourse, a dynamic managed through subtle communicative interventions (Schön, 1983).

2.2. Computational Analysis of Design Discourse

Computational approaches to design discourse have expanded from manual protocol coding (Kan and Gero, 2017) to automated text analysis. Stempfle
70 and Badke-Schaub (2002) established categories for analyzing design team communication, while Chandrasegaran et al. (2019) explored real-time visualization of verbal meeting content. Semantic distance metrics based on sentence embeddings (Reimers and Gurevych, 2019) allow quantitative evaluation of thematic distance between conversational turns. Topic modeling,
75 particularly Latent Dirichlet Allocation (LDA; Blei et al. 2003), enables quantification of topical shifts across conversational contexts.

2.3. LLMs in Creative and Collaborative Contexts

Instruction-tuned language models have been increasingly evaluated as collaborative partners (Frich et al., 2019). Lee et al. (2022) investigated
80 human-AI co-writing dynamics, and Yuan et al. (2022) examined story co-creation. In design settings, Yu et al. (2025) surveyed AI-augmented design workflows, observing that while language models rapidly generate diverse ideas, their outputs may lack situational grounding. Korre et al. (2025) reviewed LLM applications in online discussion facilitation, highlighting the
85 need for detailed behavioral profiling. Llama 3.1 8B Instruct (Dubey et al., 2024) provides an open-weight instruction-following model suitable for evaluating conversational AI behavior in research settings.

2.4. Theoretical Considerations in Human-AI Facilitation Comparison

Comparing human meeting facilitation turns with LLM-generated responses involves distinct communicative mechanisms. Bandura (1986) emphasized that human communicative repertoires emerge through observational learning, practice, and self-regulation within social environments. Facilitators in authentic settings adjust their language dynamically based on non-verbal cues, shared team history, and immediate contextual nuances (Schön, 1983). By contrast, an LLM generates text via conditional probability distributions over vocabulary tokens, conditioned on prompt context and instruction tuning, without embodied situational awareness. This structural difference suggests that LLM outputs may display high surface fluency and phase-specific vocabulary while exhibiting different semantic alignment properties relative to preceding conversational turns.

3. Method

3.1. Corpus and Data Processing

The AMI Meeting Corpus (Carletta et al., 2005) contains recordings of multi-party scenario-based meetings in which teams of four participants—a Project Manager (PM), an Industrial Designer, a User Interface Designer, and a Marketing Expert—collaboratively designed a television remote control across four meeting sessions per team.

We utilized the text transcription subset comprising 279 transcript files. Transcripts were segmented into sentence-level speaker turns, yielding 89,472 sentence-turns across 279 sessions. In the AMI scenario, the Project Manager participant holds the designated facilitation role. Non-PM speakers were categorized as design team members. Session phases were defined by relative temporal position within each transcript (first 20% Empathize/Define, middle 50% Ideate, final 30% Prototype/Evaluate), reflecting general temporal proportions in design meeting literature (Seidel and Fixson, 2013).

To extract candidate facilitation moves from Project Manager turns, we applied rule-based filters requiring: (a) word count between 8 and 60 words, (b) absence of casual non-meeting fragments, (c) presence of facilitation signals (interrogative syntax, facilitation keywords such as “what if” or “how might,” or modal verbs in turns ≥ 12 words), and (d) co-occurrence with design meeting vocabulary. This process identified 2,685 candidate facilitation

moves across 139 sessions. A stratified random sample of 199 moves, balanced across session phases, was selected as the canonical human evaluation dataset (`facilitator_moves_sampled_199.csv`).

125 To evaluate the move extraction criteria, two independent coders annotated a random subsample of 60 turns (30 accepted, 30 rejected by filters). Inter-rater agreement for facilitation move identification was Cohen’s $\kappa = 0.81$ (Cohen, 1960), indicating strong agreement (Landis and Koch, 1977).

130 3.2. LLM Inference Setup

Matched LLM facilitation responses were generated using Meta Llama 3.1 8B Instruct (Dubey et al., 2024). The model was executed in 4-bit quantized format via `bitsandbytes` (NF4 quantization, double quantization, float16 compute dtype) on an NVIDIA Tesla T4 GPU (16 GB VRAM) on Google Colab.

For each of the 199 sampled human contexts, a prompt was constructed containing:

1. System prompt: “You are an experienced design meeting facilitator... Respond with a single facilitation move of 1–3 sentences only.”
- 140 2. User prompt: The preceding 5 conversational turns (truncated to a maximum of 180 words), the explicit design phase label, and the request to provide the next facilitation move.

Responses were generated at three temperature settings ($T \in \{0.3, 0.7, 1.0\}$) with `do_sample=True`, `top_p=0.9`, and `max_new_tokens=150`, producing 597 total LLM responses (199 contexts $\times$ 3 temperatures). Quality filtering (minimum 5 words, absence of refusal phrases, $< 90\%$ n-gram context overlap) confirmed that all 597 generated responses met validation criteria.

3.3. Behavioral Dimensions and Operationalization

Each facilitation move (human PM turn or LLM response) was evaluated on six computational dimensions:

F1: Contextual Semantic Distance. Operationalized as $F_1 = 1 - \cos(\mathbf{e}_{\text{ctx}}, \mathbf{e}_{\text{resp}})$, where embeddings $\mathbf{e}$ were computed using `all-MiniLM-L6-v2` (Reimers and Gurevych, 2019). Scores lie in $[0, 2]$, where higher values represent greater embedding-space semantic distance from the preceding 5-turn context.

- F2: Directiveness (Exploratory).** Probability $P(\text{directive})$ from a fine-tuned DistilBERT classifier (Sanh et al., 2019) trained on 197 synthetic examples (99 directive, 98 open-ended; 80/20 train/test split, achieving 100% test accuracy on the synthetic split). Human validation on 100 AMI utterances yielded Cohen’s $\kappa = -0.0231$ (≈ -0.02), indicating that the synthetic model did not generalize to naturalistic discourse; directiveness is therefore treated as an exploratory, unvalidated measure.
- F3: Specificity Proxy.** Composite score averaging named-entity density (spaCy NER; Honnibal et al. 2020), Type-Token Ratio (TTR), and normalized average word length ($\min(\text{AWL}/10, 1)$).
- F4: Empathy Lexical Density.** Proportion of empathy terms (e.g., “understand,” “appreciate”) and hedging terms (e.g., “perhaps,” “might”) relative to turn word count.
- F5: Topic Divergence.** Jensen–Shannon distance (`scipy.spatial.distance.jensenshannon`) between context and response topic distributions derived from a 15-topic LDA model (Blei et al., 2003) trained on the response corpus.
- F6: Phase-Vocabulary Alignment.** Proportion of response tokens matching predefined domain vocabulary lists for Empathize/Define, Ideate, and Prototype/Evaluate phases.

3.4. Statistical Analysis

Human PM turns and LLM responses at each temperature were compared across all six dimensions using two-sided Mann-Whitney U tests (Mann and Whitney, 1947). Bonferroni correction was applied across all 18 pairwise comparisons ($6 \text{ dimensions} \times 3 \text{ temperatures}$, adjusted threshold $\alpha = 0.05/18 = 0.00278$). Effect sizes were calculated as rank-biserial correlations $r = 1 - 2U/(n_1n_2)$ (Rosenthal, 1994). The $N = 597$ LLM evaluation dataset represents 199 unique contexts evaluated across 3 decoding temperatures (199×3), rather than 597 independent contexts. Temperature evaluation accordingly functions as a repeated-context sensitivity analysis across decoding settings. Spearman rank correlations (ρ) were evaluated between temperature (0.3, 0.7, 1.0) and feature scores. Post-hoc power calculations were performed using normal approximations.

To examine sensitivity to response length differences (human $M = 23.5$ words vs. LLM $M = 38.5$ words), we conducted length-matched subsampling ($n = 72$ per group across 5 word-count quantile bins) and partial Spearman correlations controlling for word count. Non-independence across meeting sessions was evaluated using linear mixed-effects models (`statsmodels` MixedLM) with fixed effect `source` and random intercepts for `session_id`.

4. Results

4.1. Descriptive and Comparative Analysis

Table 1 summarizes descriptive metrics for the human and LLM response sets. Llama 3.1 8B Instruct responses were longer on average ($M = 38.5$ words at $T = 0.7$) than human PM turns ($M = 23.5$ words) and exhibited larger vocabulary counts across 199 turns.

Table 1: Descriptive statistics for human Project Manager turns and Llama 3.1 8B Instruct responses.

Source	N	M words	SD words	Vocab	M sentences
Human PM	199	23.5	12.7	907	1.5
LLM $T=0.3$	199	40.5	7.9	1307	1.7
LLM $T=0.7$	199	38.5	7.5	1392	1.6
LLM $T=1.0$	199	37.6	8.5	1473	1.4

Mann-Whitney U tests indicated statistically significant differences on four of the six behavioral dimensions after Bonferroni correction (Table 2). The largest effect was observed for topic divergence (Jensen–Shannon distance; $r = +0.580$, $p_{\text{Bonf}} < 0.001$), where LLM responses exhibited higher topic distance from preceding contexts than human PM turns. Human PM turns showed greater contextual semantic distance ($r = -0.537$, $p_{\text{Bonf}} < 0.001$). Llama 3.1 8B Instruct scored higher on the specificity proxy ($r = +0.254$, $p_{\text{Bonf}} < 0.001$) and phase-vocabulary alignment ($r = +0.341$, $p_{\text{Bonf}} < 0.001$).

Directiveness ($r = -0.005$, $p_{\text{Bonf}} = 1.000$, power = 0.05) and empathy lexical density ($r = -0.058$, $p_{\text{Bonf}} = 1.000$, power = 0.21) showed no statistically significant differences. Due to low post-hoc statistical power, these non-significant results cannot establish equivalence.

Table 2: Mann-Whitney U test comparisons between human PM turns and Llama 3.1 8B Instruct responses ($T = 0.7$). Rank-biserial correlation r indicates effect size (negative r denotes higher human score). p_{Bonf} corrected across 18 comparisons.

Dimension	Human Mdn	LLM Mdn	U	r	95% CI	p_{Bonf}	Power
Contextual Distance	0.742	0.593	30439	-0.537	$[-0.62, -0.45]$	<0.001	1.00
Directiveness (Exp.)	0.021	0.036	19895	-0.005	—	1.000	0.05
Specificity Proxy	0.397	0.416	14780	+0.254	$[+0.14, +0.37]$	<0.001	1.00
Empathy Lexicon	0.000	0.000	20943	-0.058	—	1.000	0.21
Topic Divergence	0.540	0.744	8309	+0.580	$[+0.48, +0.67]$	<0.001	1.00
Phase Alignment	0.000	0.000	13044	+0.341	$[+0.26, +0.43]$	<0.001	1.00

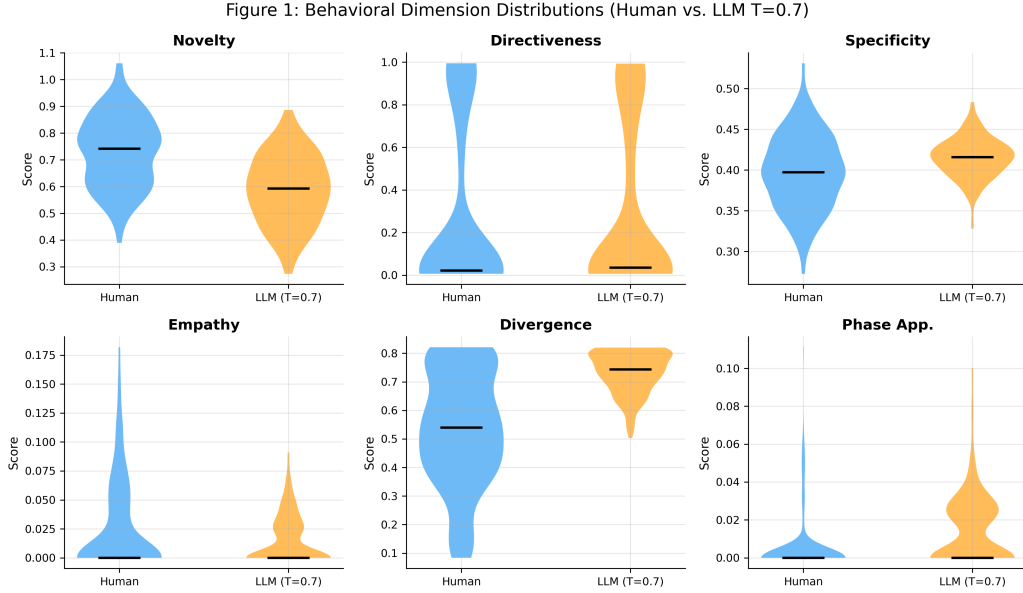


Figure 1: Violin plots showing score distributions for human Project Manager turns and Llama 3.1 8B Instruct ($T = 0.7$) responses across all six behavioral dimensions. Horizontal lines indicate medians.

4.2. Multivariate Style Space (PCA)

Principal Component Analysis on standardized features (Figure 3) showed that PC1 (22.8% variance) and PC2 (21.3% variance) together accounted for 44.1% of feature variance. PC1 loaded positively on directiveness (+0.61) and specificity (+0.34) and negatively on contextual distance (-0.49) and empathy (-0.50). PC2 loaded on topic divergence (+0.63) and phase alignment (+0.56). Human PM turns clustered toward negative PC1 values, whereas LLM responses shifted toward positive PC1 values with dispersion along PC2.

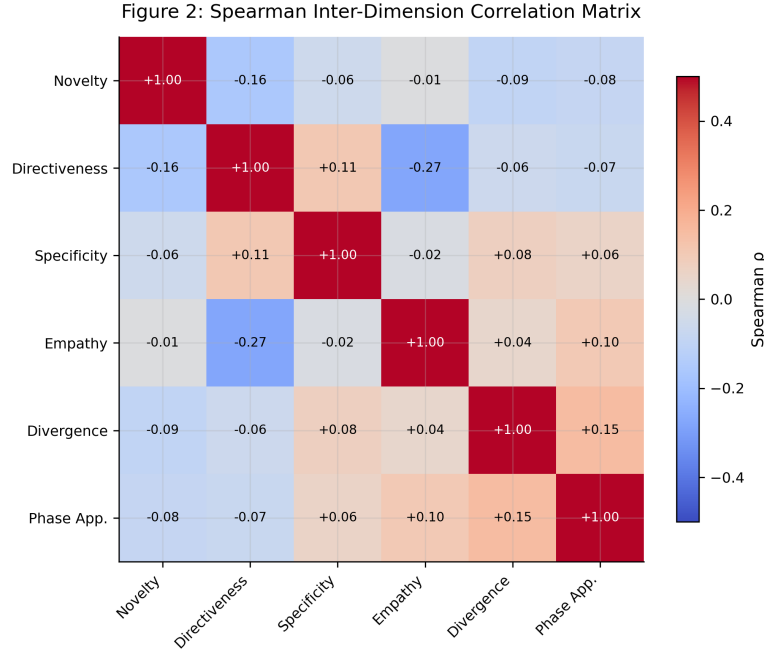


Figure 2: Spearman correlation matrix across the six behavioral dimensions ($N = 796$). Pairwise correlations remain weak ($|\rho| < 0.28$).

4.3. Temperature Sensitivity

Spearman correlation analysis across LLM temperatures (0.3, 0.7, 1.0; $N = 597$) indicated that feature scores were largely temperature-stable (Table 3). Weak significant correlations were observed for specificity ($\rho = +0.116$, $p = 0.005$) and phase-vocabulary alignment ($\rho = -0.102$, $p = 0.013$).

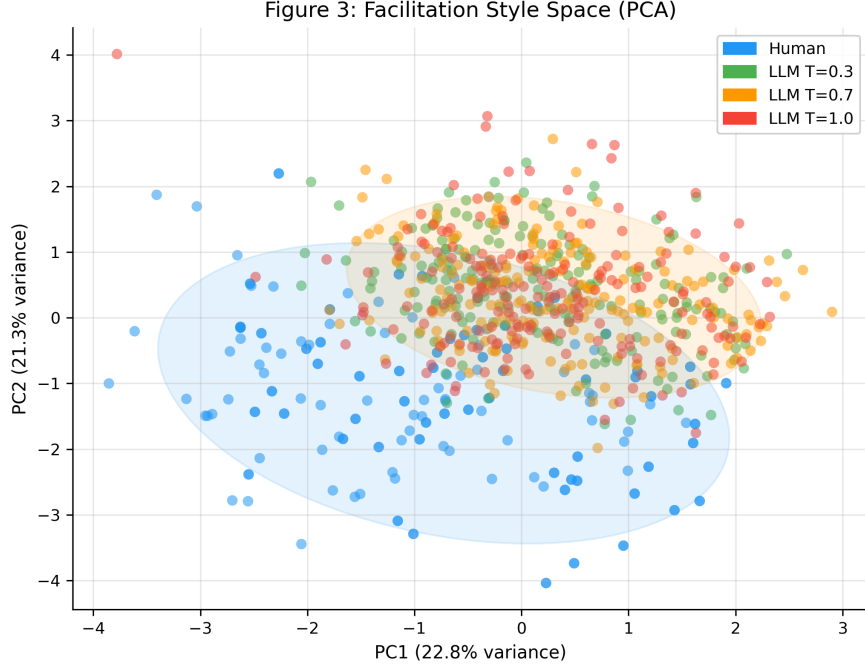


Figure 3: Principal Component Analysis (PCA) projection of the six-dimensional behavioral feature space for human PM turns and Llama 3.1 8B Instruct responses.

Table 3: Spearman correlations between decoding temperature ($T \in \{0.3, 0.7, 1.0\}$) and feature scores across Llama 3.1 8B Instruct responses ($N = 597$).

Dimension	ρ	p	Sig.
Contextual Distance	-0.012	0.772	—
Directiveness (Exp.)	-0.010	0.804	—
Specificity Proxy	+0.116	0.005	**
Empathy Lexicon	+0.005	0.903	—
Topic Divergence	-0.056	0.170	—
Phase Alignment	-0.102	0.013	*

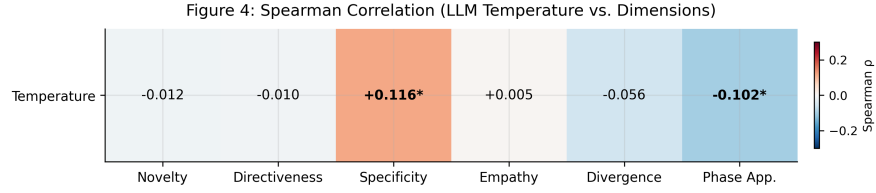


Figure 4: Heatmap of median feature scores across human PM turns and Llama 3.1 8B Instruct temperatures ($T = 0.3, 0.7, 1.0$).

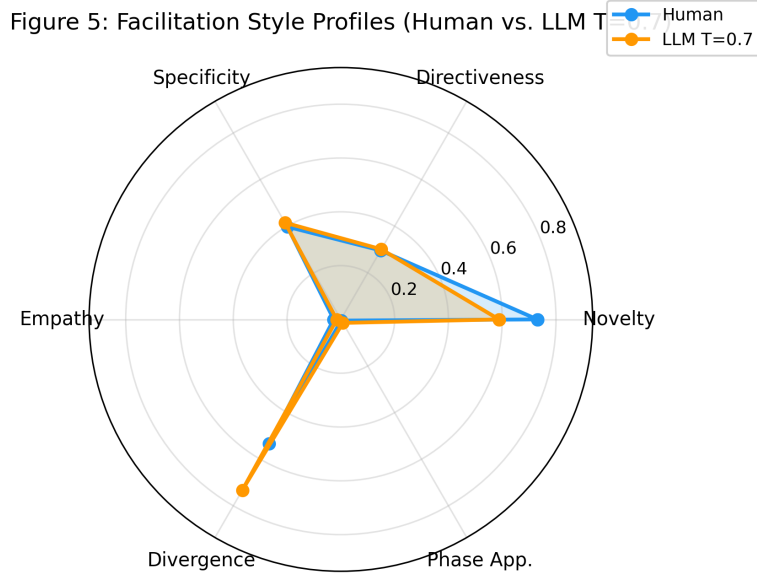


Figure 5: Radar chart comparing normalized median behavioral profiles of human PM turns and Llama 3.1 8B Instruct responses across the six dimensions.

4.4. Sensitivity and Robustness Analyses

Sensitivity analyses evaluating response length and session clustering remained qualitatively consistent with the primary findings (Table A.4). In
length-matched subsampling ($n = 72$ per group), differences remained sig-
nificant for contextual distance ($r = -0.309$), topic divergence ($r = +0.499$),
specificity proxy ($r = +0.706$), and phase alignment ($r = +0.281$). Partial
Spearman correlations controlling for word count confirmed these directions.
Linear mixed-effects models with session random intercepts confirmed fixed-
effect differences for contextual distance ($\beta = -0.1426$, $p < 0.001$), topic
divergence ($\beta = +0.1993$, $p < 0.001$), specificity ($\beta = +0.0172$, $p < 0.001$),
and phase alignment ($\beta = +0.0100$, $p < 0.001$).

5. Discussion

5.1. Interpreting Behavioral Differences

The empirical comparison indicates distinct output properties between
Llama 3.1 8B Instruct responses and human Project Manager turns in the
AMI design scenario. The higher topic divergence ($r = +0.580$) observed in
LLM responses indicates that generated moves introduce thematic content
that is further from the immediate 5-turn context in LDA topic space. Hu-
man PM turns, by contrast, displayed higher embedding-space contextual
distance ($r = -0.537$), reflecting distinct phrasing while remaining topic-
bound.

These patterns align with structural generation differences. Llama 3.1
8B Instruct conditions its responses on prompt instructions and general pre-
training patterns, frequently producing structured coaching prompts (e.g.,
“Let’s take a step back and ask...”). Human PM participants in naturalistic
meeting settings respond with conversational turn-taking, disfluencies, and
immediate context references.

5.2. Methodological Limitations and Constraints

Several limitations constrain the interpretation of these findings:

1. **Single-Model Evaluation:** Evaluation was conducted exclusively on Llama 3.1 8B Instruct (4-bit quantized). Behavior may differ across larger architectures or alternative model families.

- 260 2. **Scenario-Based Participant Corpus:** Human turns were drawn from Project Manager participants in the AMI remote-control design scenario, which may not represent professional design facilitators in industry settings.
- 265 3. **Exploratory Directiveness Instrument:** The DistilBERT directiveness classifier achieved high accuracy on synthetic test data but failed human validation on 100 AMI utterances ($\kappa = -0.0231$). Directiveness results are exploratory and unvalidated.
- 270 4. **Prompting and Phase Conditioning:** The design phase label was explicitly provided in the LLM prompt, and phase assigned by temporal position. Phase-vocabulary alignment measures vocabulary match rather than independent phase inference.
5. **Linguistic Proxies:** Specificity and empathy measures rely on surface lexical proxies (NER/TTR/word length and lexicon density) rather than direct psychological measurements.
- 275 6. **Statistical Power on Null Findings:** Non-significant findings for directiveness ($p_{\text{Bonf}} = 1.000$, power = 0.05) and empathy ($p_{\text{Bonf}} = 1.000$, power = 0.21) reflect insufficient power and cannot demonstrate equivalence.
- 280 7. **Independent-Sample Inferential Assumption:** The 199 LLM responses at each temperature are context-matched to the 199 human PM turns, and the three temperatures reuse the same 199 contexts. However, the primary inferential analyses utilize independent-sample Mann-Whitney U tests. The tests do not model paired or hierarchical error structures and should be interpreted as independent-sample comparisons rather than fully paired hierarchical inference.

285 6. Conclusion

This study compared Llama 3.1 8B Instruct responses with naturally occurring human Project Manager turns from the AMI Meeting Corpus across six computational behavioral dimensions. Statistically significant differences were observed on four dimensions: Llama 3.1 8B Instruct generated responses with higher topic divergence ($r = +0.58$), higher specificity proxy ($r = +0.25$), and higher phase-vocabulary alignment ($r = +0.34$), whereas human PM turns exhibited greater contextual semantic distance ($r = -0.54$). Directiveness and empathy showed no significant differences under low statistical power. These results document baseline behavioral characteristics of

295 Llama 3.1 8B Instruct in meeting facilitation contexts and highlight methodological considerations for computational facilitation research.

Declaration of Interest

The authors declare no conflict of interest.

Data Availability

300 The AMI Meeting Corpus is available at <https://groups.inf.ed.ac.uk/ami/corpus/>. Analysis scripts, processed data, and validation harnesses are available at <https://github.com/Ayaan577/llm-facilitation-behavior>.

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Appendix A. Robustness Check Results

Table A.4: Robustness Checks—Length-Matched Subsampling, Partial Correlations, and MixedLM Results (Human PM vs. LLM $T = 0.7$). Original r : rank-biserial from Mann-Whitney U . Length-Matched r : rank-biserial after quantile bin matching ($n = 72$). Partial ρ : Spearman partial correlation controlling for word count. ME β : fixed-effect estimate from linear mixed-effects model with session random intercepts.

Dimension	Original r	Length-Matched r	Partial ρ	ME β	p_{Bonf} (LMM)	Qua
Contextual Distance	−0.54	−0.31	−0.23	−0.143	<0.001	
Topic Divergence	+0.58	+0.50	+0.41	+0.199	<0.001	
Specificity Proxy	+0.25	+0.71	+0.51	+0.017	<0.001	
Phase Alignment	+0.34	+0.28	+0.18	+0.010	<0.001	
Empathy Lexicon	−0.06	−0.15	−0.20	−0.011	0.005	
Directiveness (Exp.)	−0.005	−0.18	−0.11	+0.005	1.000	